



Name of Student: Anurag Thakur	PRN:25070122040
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 16	
TITLE : User-Defined Exceptions	

Problem Statement

Write a Java program to create and handle a user-defined exception.

Learning Objectives

To create and implement custom exceptions for application-specific error handling.

Theory:

User-defined exceptions are created by extending the Exception class. They allow developers to define meaningful exceptions for specific business logic. Custom exceptions improve program readability and simplify debugging by providing descriptive error messages. They are widely used in enterprise software and validation systems.

Programme

```
import java.util.Scanner;

class InvalidAgeException extends Exception {
    public InvalidAgeException(String message) {
        super(message);
    }
}

public class VotingSystem {

    static void checkAge(int age) throws InvalidAgeException {
        if (age < 18) {
            throw new InvalidAgeException(
                "You are not eligible for voting."
            );
        }
    }
}
```

```

    );
}

    System.out.println("You are eligible for voting.");
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    try {
        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        checkAge(age);
    }
    catch (InvalidAgeException e) {
        System.out.println("Error: " + e.getMessage());
    }

    sc.close();
}
}

```

```

import java.util.Scanner;

class UnderAgeException extends Exception {
    public UnderAgeException(String message) {
        super(message);
    }
}

public class DrivingLicenseSystem {

```

```
static void checkEligibility(int age) throws UnderAgeException {
    if (age < 18) {
        throw new UnderAgeException(
            "Age must be 18 or above to apply for a driving license."
        );
    }

    System.out.println("You are eligible for a driving license.");
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    try {
        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        checkEligibility(age);
    }
    catch (UnderAgeException e) {
        System.out.println("Error: " + e.getMessage());
    }

    sc.close();
}
```

Output

```
Problems  Output  Debug Console  Terminal  Ports

cd "/Users/rakeshkumarchaudhary/Experiment 16"
cd "/Users/rakeshkumarchaudhary/Experiment 16/" && javac VotingSystem.j
● rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 16 % cd "/Users/rak
● <s/rakeshkumarchaudhary/Experiment 16/" && javac VotingSystem.java && j
Enter your age: 12
Error: You are not eligible for voting.
○ rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 16 % □
```

```
Problems  Output  Debug Console  Terminal  Ports

● rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 16 % cd "/Users/rakeshkuma
● rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 16 % cd "/Users/rakeshkuma
tem.java && java DrivingLicenseSystem
Enter your age: 17
Error: Age must be 18 or above to apply for a driving license.
○ rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 16 % □
```

Conclusion

The program successfully demonstrates the creation and handling of user-defined exceptions to manage application-specific validation errors effectively.